

Tetration, perfect powers, and frozen decimal digits

A closed formula for the smallest n th perfect power
whose tetration has congruence speed n

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Abstract. Let $V(a)$ be the eventual number of additional decimal digits shared by successive power towers with integer base $a > 1$, where $10 \nmid a$. A 2025 MathOverflow question asks for the smallest n th perfect power $\rho(n) = x^n > 1$ with $V(x^n) = n$, and proposes that the minimizer always ends in 5 once $n \geq 3$. We give a complete elementary proof of that pattern and an explicit formula. Set $s = n - \nu_2(n)$. Then $\rho(1) = 2$, $\rho(2) = 49$, and, for $n \geq 3$,

$$\rho(n) = B_s^n, \quad B_s = \begin{cases} 2^s - 1, & s \equiv 0 \pmod{4}, \\ 3 \cdot 2^s - 1, & s \equiv 1 \pmod{4}, \\ 2^s + 1, & s \equiv 2 \pmod{4}, \\ 3 \cdot 2^s + 1, & s \equiv 3 \pmod{4}. \end{cases}$$

The key is a deterministic size barrier: every competing root not divisible by 5 must satisfy $x^2 + 1 \geq 5^{n-\nu_5(n)}$. This excludes all competitors for $n \geq 25$; five small Hensel-lifting certificates settle the remaining exceptional comparisons. We also solve the stricter maximal-degree version: only $n = 3$ changes, from 25^3 to 55^3 . Further results include the exact choice between the two conjectured decimal tails, a two-parameter extremal theorem, density formulas, and asymptotics. The archive contains standard-library Python code, exact certificates, and reproducible tests.

Scope and status. The argument below is a complete proof for the precisely stated arithmetic problem, not a numerical extrapolation. The foundational congruence-speed formula is prior work and is rederived for completeness. The consulted question leaves the all- n extremal pattern conjectural; an exhaustive claim of bibliographic priority is not made. This report has not undergone external peer review or proof-assistant verification.

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1 The problem, its provenance, and the answer

1.1 Why this particular question?

An integer power tower can be far too large to write down while its last few digits are easy to study. This makes modular tetration a useful place to seek a tractable problem: one can replace the enormous integers by small local invariants, then turn a computational pattern into a proof. The relevant objects here are ordinary integer tetrations $a \uparrow\uparrow h$, not analytic interpolations in the height.

The eventual decimal congruence-speed formula was established in work of Ripà [1, 2]; stable-digit counts were subsequently studied by Ripà and Onnis [3]. Those results are not the open problem attacked here. The later extremal question asks that the base itself be an n th perfect power and that its speed equal the same integer n .

Specifically, Ripà's February 2025 MathOverflow question asks for a closed formula for this minimum. Alekseyev's answer gives 50 root values and observes the apparent predominance of the final-digit-5 case. The question also conjectures two fixed ten-digit tails for all $n > 10$ [4]. A related paper, published in 2024 and deposited on arXiv in January 2026, constructs perfect powers with prescribed speeds. Existence is different from the global minimality proved here [5].

The choice is deliberately narrow. No conclusion below is a proposed resolution of analytic tetration, arbitrary hyperoperation interpolation, or ordinal fast-growing-hierarchy questions. It is a concrete extremal problem inside the arithmetic of tetration.

1.2 Definitions and conventions

For $a \geq 2$, put

$$T_0(a) = 1, \quad T_{h+1}(a) = a^{T_h(a)} \quad (h \geq 0). \quad (1)$$

Thus $T_h(a) = a \uparrow\uparrow h$. For a nonzero integer u and a prime p , $\nu_p(u)$ is the largest $j \geq 0$ such that $p^j \mid u$. Define

$$\Delta_h(a) = T_{h+1}(a) - T_h(a), \quad D_h(a) = \min\{\nu_2(\Delta_h(a)), \nu_5(\Delta_h(a))\}. \quad (2)$$

Exactly $D_h(a)$ trailing decimal positions agree between these two towers. For $10 \nmid a$, we shall prove that

$$D_h(a) - D_{h-1}(a)$$

is eventually a positive constant; this constant is $V(a)$. For convenience only, set $V(1) = 0$. Multiples of 10 are excluded from the extremal problem because their speed is not eventually a finite constant.

Definition 1.1. For $n \geq 1$, define

$$r(n) = \min\{x \in \mathbb{N} : x > 1, 10 \nmid x, V(x^n) = n\}, \quad (3)$$

$$\rho(n) = r(n)^n. \quad (4)$$

The minimum will be shown to exist. Since $x \mapsto x^n$ is strictly increasing on the positive integers, minimizing x and minimizing x^n are equivalent.

An n th perfect power need not have *maximal* perfect-power degree n . For example, $25^3 = 5^6$ is an admissible cube under (4). We solve the maximal-degree variant separately in Section 7 rather than silently changing the question.

1.3 Main theorem

Theorem 1.2 (Closed formula). *We have $r(1) = 2$ and $r(2) = 7$. For every $n \geq 3$, let*

$$s = n - \nu_2(n) \geq 2.$$

Then

$$r(n) = B_s, \quad B_s = \begin{cases} 2^s - 1, & s \equiv 0 \pmod{4}, \\ 3 \cdot 2^s - 1, & s \equiv 1 \pmod{4}, \\ 2^s + 1, & s \equiv 2 \pmod{4}, \\ 3 \cdot 2^s + 1, & s \equiv 3 \pmod{4}. \end{cases} \quad (5)$$

Consequently $r(n) \equiv \rho(n) \equiv 5 \pmod{10}$ for all $n \geq 3$.

The first root values are

$$2, 7, 25, 5, 95, 95, 385, 95, 1535, 1535, 6145, 1025, \dots$$

The proof has two genuinely different parts. Finding the smallest admissible multiple of 5 is an elementary congruence calculation. Proving that it beats *every* other residue class requires the size barrier in Section 5. Checking many initial terms would not replace that second part.

2 From power towers to two local valuations

We give the details so that the main theorem does not rely on treating a large published case distinction as a black box. The resulting compact formula is equivalent to the known decimal congruence-speed formula [2, 3].

2.1 The elementary lifting facts

Lemma 2.1 (Lifting near 1). *Let $u > 1$ and $m \geq 1$ be integers. If $u \equiv 1 \pmod{5}$, then*

$$\nu_5(u^m - 1) = \nu_5(u - 1) + \nu_5(m). \quad (6)$$

If $u \equiv 1 \pmod{4}$, then

$$\nu_2(u^m - 1) = \nu_2(u - 1) + \nu_2(m). \quad (7)$$

In addition, if u is odd and m is even, then

$$\nu_2(u^m - 1) = \nu_2(u - 1) + \nu_2(u + 1) + \nu_2(m) - 1. \quad (8)$$

Proof. For either prime p , if $p \nmid q$ and $u \equiv 1 \pmod{p}$, the factorization

$$\frac{u^q - 1}{u - 1} = 1 + u + \dots + u^{q-1} \equiv q \not\equiv 0 \pmod{p}$$

shows that raising to the q th power leaves the valuation unchanged. It therefore suffices to understand raising to the p th power.

For $p = 5$, write $u = 1 + 5^j w$ with $5 \nmid w$ and $j \geq 1$. In the binomial expansion of $u^5 - 1$, the first term has valuation $j + 1$, whereas each other term has larger valuation. Thus $\nu_5(u^5 - 1) = j + 1$. Repeating this step proves (6). For $p = 2$ with $u \equiv 1 \pmod{4}$, the factorization $u^2 - 1 = (u - 1)(u + 1)$ gives $\nu_2(u + 1) = 1$, so each squaring increases the valuation by exactly one. This proves (7). Finally, apply (7) to $u^2 \equiv 1 \pmod{8}$ and exponent $m/2$:

$$\nu_2(u^m - 1) = \nu_2(u^2 - 1) + \nu_2(m/2),$$

which is (8). □

For odd $a > 1$ and, respectively, $5 \nmid a > 1$, define

$$c_2(a) = \nu_2(a^2 - 1) - 1, \quad c_5(a) = \nu_5(a^4 - 1). \quad (9)$$

For odd a , one of $a - 1, a + 1$ has valuation exactly one at 2, so

$$c_2(a) = \max\{\nu_2(a - 1), \nu_2(a + 1)\} \geq 2. \quad (10)$$

The functions c_2, c_5 are only used on the domains just specified. In particular, no valuation of zero is needed.

2.2 The recurrence for tower differences

For $h \geq 1$, direct factorization of (1) gives

$$\Delta_h = a^{T_{h-1}}(a^{\Delta_{h-1}} - 1). \quad (11)$$

If a prime p divides a , the second factor is a p -adic unit. Hence

$$\nu_p(\Delta_h) = \nu_p(a)T_{h-1} \quad \text{when } p \mid a. \quad (12)$$

This term eventually dominates any affine function of h . Indeed, $T_h \geq 2^h$: for an integer $u \geq 1$, $2^u \geq 2u$, and induction in (1) gives $T_{h+1} \geq 2T_h$.

If a is odd, all T_h are odd, so every Δ_h is even. By Lemma 2.1, (11) then implies

$$\nu_2(\Delta_h) = \nu_2(\Delta_{h-1}) + c_2(a) = \nu_2(a - 1) + hc_2(a). \quad (13)$$

This identity holds for every $h \geq 0$.

Suppose instead that $5 \nmid a$. Once $h \geq 3$, we have $4 \mid \Delta_{h-1}$. To see this when a is odd, use (13) and $c_2(a) \geq 2$. When a is even, T_j is divisible by 4 for every $j \geq 2$, and so are the corresponding differences. Now apply (6) to $u = a^4$ and $m = \Delta_{h-1}/4$:

$$\nu_5(\Delta_h) = \nu_5(\Delta_{h-1}) + c_5(a) \quad (h \geq 3). \quad (14)$$

Thus the 5-adic valuations are eventually affine with slope $c_5(a)$.

Theorem 2.2 (Compact congruence-speed formula). *For $a > 1$ and $10 \nmid a$, the constant $V(a)$ exists and satisfies*

$$V(a) = \begin{cases} c_5(a), & 2 \mid a, \\ c_2(a), & 5 \mid a, \\ \min\{c_2(a), c_5(a)\}, & \gcd(a, 10) = 1. \end{cases} \quad (15)$$

Proof. When a is even, the 2-adic term in D_h has the rapidly growing form (12), while the 5-adic term is eventually affine with slope $c_5(a)$. Their minimum therefore eventually equals the latter. The case $5 \mid a$ is symmetric, with slope $c_2(a)$. When a is coprime to 10, both terms are eventually affine. The minimum of two affine functions is eventually affine with slope the smaller of the two slopes; if the slopes are equal, only the intercepts matter. This proves (15) and existence of $V(a)$. $\square$

Remark 2.3 (Transient behavior is not ignored). The theorem concerns eventual slopes, not every height. For example, for $a = 5$ the first values D_0, D_1, D_2, D_3, D_4 are 0, 1, 5, 8, 10; only later is each increment 2. For $a = 807$, the minimum switches between local affine contributions before settling at slope 3. The verification code checks tower residues directly, rather than assuming that a low-height observed difference must equal $V(a)$.

3 Perfect powers shift the local speeds

Proposition 3.1 (Power-transfer law). *For $x > 1$, $10 \nmid x$, and $k \geq 1$,*

$$V(x^k) = \begin{cases} c_5(x) + \nu_5(k), & 2 \mid x, \\ c_2(x) + \nu_2(k), & 5 \mid x, \\ \min\{c_2(x) + \nu_2(k), c_5(x) + \nu_5(k)\}, & \gcd(x, 10) = 1. \end{cases} \quad (16)$$

Proof. For odd x , apply (7) to $u = x^2$ to obtain

$$c_2(x^k) = \nu_2(x^{2k} - 1) - 1 = \nu_2(x^2 - 1) - 1 + \nu_2(k) = c_2(x) + \nu_2(k).$$

For $5 \nmid x$, apply (6) to $u = x^4$:

$$c_5(x^k) = \nu_5(x^{4k} - 1) = c_5(x) + \nu_5(k).$$

Substitute into Theorem 2.2. □

This identity is the main computational simplification. Evaluating $V(x^k)$ requires no k th power, much less any tower with base x^k . Only the valuations of $x \pm 1$, $x^2 + 1$, and k enter.

For the diagonal problem $k = n$, write

$$s = n - \nu_2(n), \quad t = n - \nu_5(n). \quad (17)$$

If $5 \mid x$ and x is odd, then

$$V(x^n) = n \iff c_2(x) = s. \quad (18)$$

On the other hand, every admissible root with $5 \nmid x$ necessarily satisfies

$$c_5(x) \geq t. \quad (19)$$

The second assertion holds whether x is even or odd. For odd x it is only a necessary condition, which is enough for a lower bound.

For $n \geq 3$, $s \geq 2$. This follows immediately for odd n ; for even $n = 2^j u$ with u odd, $n - j \geq 2$, including the smallest case $n = 4$. Also $t \geq 1$ for all positive n .

4 The smallest root divisible by five

Lemma 4.1 (Exact 2-adic congruence classes). *For an odd $x > 1$ and $s \geq 2$,*

$$c_2(x) = s \iff x \equiv 2^s - 1 \text{ or } 2^s + 1 \pmod{2^{s+1}}. \quad (20)$$

Proof. By (10), the larger of the valuations of $x - 1, x + 1$ must be exactly s . This is equivalent to one of those two integers being 2^s times an odd number. The other has valuation one. This yields exactly the two stated classes. □

Proposition 4.2 (Restricted minimum). *For $s \geq 2$, the least odd positive multiple of 5 satisfying $c_2(x) = s$ is B_s from (5). In particular,*

$$5 \mid B_s, \quad c_2(B_s) = s, \quad 2^s - 1 \leq B_s \leq 3 \cdot 2^s + 1. \quad (21)$$

Proof. By Lemma 4.1, every candidate has the form $x = j2^s + \varepsilon$, where $j \geq 1$ is odd and $\varepsilon \in \{-1, 1\}$. The powers of 2 modulo 5 have period four. The smallest admissible choices are therefore

$s \bmod 4$	$2^s \bmod 5$	smallest odd j	ε	B_s
0	1	1	-1	$2^s - 1$
1	2	3	-1	$3 \cdot 2^s - 1$
2	4	1	+1	$2^s + 1$
3	3	3	+1	$3 \cdot 2^s + 1$

In the second and fourth rows, neither sign works for $j = 1$, and $j = 2$ is forbidden because the valuation must be exact. In all rows the listed j works; every larger eligible odd j produces a larger x . The valuation and bounds follow directly. $\square$

For every $n \geq 3$, this constructs an admissible n th power B_s^n and proves that no smaller multiple-of-5 root can work. There remains a global question: could a root in a different decimal class be even smaller?

5 A deterministic barrier against all other roots

Lemma 5.1 (Quadratic size barrier). *Let $x > 1$, $5 \nmid x$, and $t \geq 1$. If $c_5(x) \geq t$, then*

$$x^2 + 1 \geq 5^t. \quad (22)$$

Proof. Factor

$$x^4 - 1 = (x - 1)(x + 1)(x^2 + 1).$$

Exactly one of the three factors is divisible by 5. Indeed, $x \equiv 1, -1, 2, -2 \pmod{5}$ determines which factor it is, and two of these factors cannot simultaneously be divisible by 5. Thus all the factor 5^t divides one of them. Each is positive and at most $x^2 + 1$ when $x \geq 2$, proving the result. $\square$

The factorization is decisive. Merely using $5^t \mid x^4 - 1$ would give only a fourth-root lower bound, too weak to beat a candidate of order 2^n . Isolating the factor $x^2 + 1$ raises the lower bound to approximately $5^{n/2}$, which eventually dominates 2^n .

Proposition 5.2 (A sufficient global-minimality condition). *For $n \geq 3$, let s, t be as in (17). If*

$$5^t > B_s^2 + 1, \quad (23)$$

then $r(n) = B_s$.

Proof. The root B_s is admissible and is least among roots divisible by 5. A competing root $x \leq B_s$ not divisible by 5 would have $c_5(x) \geq t$ by (19). But Lemma 5.1 would imply $5^t \leq x^2 + 1 \leq B_s^2 + 1$, a contradiction. $\square$

Lemma 5.3 (Uniform cutoff). *Condition (23) holds for every $n \geq 25$.*

Proof. Since $5^{\nu_5(n)} \leq n$ and $s \leq n$,

$$5^t \geq \frac{5^n}{n}, \quad B_s^2 + 1 \leq (3 \cdot 2^n + 1)^2 + 1.$$

Put $H_n = (3 \cdot 2^n + 1)^2 + 1$. Direct expansion gives

$$H_{n+1} = 36 \cdot 4^n + 12 \cdot 2^n + 2 < 4H_n.$$

Consequently, for $n \geq 5$,

$$\frac{5^{n+1}/((n+1)H_{n+1})}{5^n/(nH_n)} > \frac{5n}{4(n+1)} > 1.$$

The starting inequality at $n = 25$ is the exact integer calculation

$$\begin{aligned} 5^{25} - 25H_{25} &= 298023223876953125 - 253327484072755250 \\ &= 44695739804197875 > 0. \end{aligned}$$

Thus $5^n/n > H_n$ for every $n \geq 25$, proving the claim. $\square$

Notice that no assumption about random 5-adic digits occurs here. The lower bound remains valid even if the residues of square roots of -1 have extremely atypical digit patterns.

6 The finite range: exact, inspectable certificates

6.1 The routine comparisons

The uniform cutoff leaves only $3 \leq n \leq 24$. In Table 1, $Q_n = \lfloor 5^t/(B_s^2 + 1) \rfloor$. A positive Q_n certifies (23): equality $5^t = B_s^2 + 1$ cannot occur, because $5 \nmid B_s$.

n	s	t	B_s	Q_n
3	3	3	25	0
4	2	4	5	24
5	5	4	95	0
6	5	6	95	1
7	7	7	385	0
8	5	8	95	43
9	9	9	1535	0
10	9	9	1535	0
11	11	11	6145	1
12	10	12	1025	232
13	13	13	24575	2
14	13	14	24575	10
15	15	14	98305	0
16	12	16	4095	9099
17	17	17	393215	4
18	17	18	393215	24
19	19	19	1572865	7
20	18	19	262145	277
21	21	21	6291455	12
22	21	22	6291455	60
23	23	23	25165825	18
24	21	24	6291455	1505

Table 1: The finite comparison range. Only $n = 3, 5, 7, 9, 10, 15$ require more than the quadratic size barrier. All entries are exact integers.

6.2 A two-root lifting lemma

Lemma 6.1. *For every $k \geq 1$, $z^2 \equiv -1 \pmod{5^k}$ has exactly two roots. Starting with either root modulo 5, each root has exactly one lift to a root modulo 5^{k+1} . The two roots are negatives of each other.*

Proof. Modulo 5, the roots are 2 and 3. Suppose $r^2 + 1 = 5^k q$. Every lift is $r + j5^k$, with $j \in \{0, 1, 2, 3, 4\}$. Modulo 5^{k+1} ,

$$(r + j5^k)^2 + 1 \equiv 5^k(q + 2rj).$$

Since $5 \nmid 2r$, there is a unique j modulo 5 satisfying $q + 2rj \equiv 0 \pmod{5}$. Every root at the next level reduces to a root at the previous level, so no others appear. Negation interchanges the two initial roots and preserves the equation at every level. $\square$

For a root $x > 1$ with $c_5(x) \geq t \geq k$, the factorization in Lemma 5.1 gives one of

$$x \equiv 1, -1 \pmod{5^k}, \quad x^2 \equiv -1 \pmod{5^k}. \quad (24)$$

The smallest permitted $x > 1$ in the first pair of classes is $5^k - 1$. If d_k is the smaller root of -1 modulo 5^k , the second pair requires $x \geq d_k$. Thus

$$x \geq \min\{5^k - 1, d_k\}. \quad (25)$$

6.3 Five certificates settle six indices

The following table is enough; it does not search through a large interval of candidate roots.

n	B_s	k	5^k	d_k	$(d_k^2 + 1)/5^k$
3	25	3	125	57	26
5	95	4	625	182	53
7	385	5	3125	1068	365
9,10	1535	7	78125	32318	13369
15	98305	8	390625	110443	31226

Table 2: Certificates for the exceptional comparisons. In each row, $k \leq t$, $B_s < d_k < 5^k/2$, and $B_s < 5^k - 1$.

Each quotient in the last column is integral, so d_k is a root of -1 modulo 5^k . It is less than half the modulus, and Lemma 6.1 says the only other root is $5^k - d_k$. Therefore it really is the smaller root. The inequalities in the caption, combined with (25), exclude every root $x \leq B_s$ not divisible by 5 in each exceptional case.

Completion of Theorem 1.2. For $n \geq 25$, use Lemma 5.3 and Proposition 5.2. For $3 \leq n \leq 24$, use Table 1 when $Q_n > 0$ and Table 2 otherwise. The restricted minimum is always supplied by Proposition 4.2. For $n = 1$, $V(2) = \nu_5(15) = 1$, so the smallest possible root is 2. For $n = 2$, direct application of (16) gives

$$\begin{array}{c|cccccc} x & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline V(x^2) & 1 & 1 & 1 & 3 & 1 & 2. \end{array}$$

Hence $r(2) = 7$. Finally B_s is odd and divisible by 5, and its positive powers all end in 5. $\square$

7 The stricter maximal-degree problem

The original discussion also asks about powers of *exact* degree n , meaning that the answer cannot also be a d th power with $d > n$ [4]. To avoid ambiguity, define the maximal perfect-power degree of an integer $a > 1$ to be

$$\text{ppdeg}(a) = \max\{d \geq 1 : a = y^d \text{ for an integer } y > 1\}.$$

If $a = \prod p^{e_p}$, this maximum is $\gcd_p e_p$. Consequently,

$$\text{ppdeg}(x^n) = n \iff x \text{ is not a perfect power of degree } > 1. \quad (26)$$

Let $\rho_{\text{exact}}(n)$ be the minimum subject to $V(a) = n$ and $\text{ppdeg}(a) = n$, with $10 \nmid a$.

Lemma 7.1 (The restricted roots are almost never perfect powers). *For $s \geq 2$, B_s is a perfect power of degree greater than one if and only if $s = 3$. The exceptional value is $B_3 = 25$.*

Proof. Any perfect power is either a square or a q th power for an odd prime q . Its root is odd because B_s is odd.

First consider $B_s = 2^s - 1$, where $s \equiv 0 \pmod{4}$. If $B_s = y^2$, then $y^2 + 1 = 2^s$, impossible modulo 8 for $s \geq 2$. If $B_s = y^q$ with odd $q \geq 3$, factor $y^q + 1$ as $(y + 1)$ times

$$y^{q-1} - y^{q-2} + \cdots - y + 1.$$

This second factor is odd and greater than one, so their product cannot be a power of 2.

Next consider $B_s = 2^s + 1$, where $s \equiv 2 \pmod{4}$. For an odd exponent, $(y^q - 1)/(y - 1)$ is odd and greater than one, again contradicting a power of 2. For a square, $(y - 1)(y + 1) = 2^s$ consists of two powers of 2 differing by 2. They must be 2 and 4, giving $s = 3$, outside this case.

Suppose $B_s = 3 \cdot 2^s - 1$, where $s \equiv 1 \pmod{4}$ and $s \geq 5$. A square would give $y^2 + 1 = 3 \cdot 2^s$, impossible modulo 8. For an odd prime exponent, the odd factor

$$y^{q-1} - y^{q-2} + \cdots - y + 1$$

would divide 3. But it is at least $y^2 - y + 1 \geq 7$ for odd $y \geq 3$. This is impossible.

Finally suppose $B_s = 3 \cdot 2^s + 1$, where $s \equiv 3 \pmod{4}$. For an odd prime exponent, the factor $(y^q - 1)/(y - 1)$ is odd, divides 3, and is at least $y^2 + y + 1 \geq 13$, a contradiction. For a square, put $u = (y - 1)/2$ and $v = (y + 1)/2$. Then

$$v - u = 1, \quad \gcd(u, v) = 1, \quad uv = 3 \cdot 2^{s-2}.$$

The odd member of the consecutive pair is either 1 or 3. The pair 1, 2 has product not divisible by 3. The remaining pairs are 2, 3 and 3, 4, giving $s = 3$ and $s = 4$, respectively. The congruence $s \equiv 3 \pmod{4}$ leaves only $s = 3$, with $B_s = 25$. $\square$

Theorem 7.2 (Exact-degree minimum). *For every $n \geq 1$,*

$$\rho_{\text{exact}}(n) = \begin{cases} 55^3 = 166375, & n = 3, \\ \rho(n), & n \neq 3. \end{cases} \quad (27)$$

Proof. The roots $r(1) = 2, r(2) = 7$ are not perfect powers. For $n \geq 3$, Lemma 7.1 says that only $s = n - \nu_2(n) = 3$ can cause trouble. Writing $j = \nu_2(n)$, this equation gives $n = j + 3$.

For $j \geq 3$, $2^j > j + 3$, impossible because $2^j \mid n$. The cases $j = 0, 1, 2$ leave only $n = 3$. Thus for $n \neq 3$, the unrestricted minimizer already has maximal degree n .

For $n = 3$, $c_2(55) = 3$, so $V(55^3) = 3$. The root $55 = 5 \cdot 11$ is not a perfect power. Among smaller odd multiples of 5, the only one with $c_2(x) = 3$ is 25, as can be checked from the list 5, 15, 25, 35, 45 or from (20). It is a square. Every root not divisible by 5 with $c_5(x) \geq 3$ is at least 57 by (25) at $k = 3$. Therefore no root below 55 satisfies both required conditions. $\square$

This strengthening needs no appeal to a deep theorem about consecutive perfect powers. The four elementary factorizations above are sufficient.

8 The decimal-tail conjecture, with the sign determined

For $d \geq 1$, let E_d be the unique residue in $[0, 10^d)$ satisfying

$$E_d \equiv 0 \pmod{5^d}, \quad E_d \equiv 1 \pmod{2^d}. \quad (28)$$

The Chinese remainder theorem gives existence and uniqueness, as well as the explicit finite formula

$$E_d = 5^d((5^d)^{-1} \bmod 2^d). \quad (29)$$

Here the inverse is represented in $\{0, \dots, 2^d - 1\}$. Since $E_d^2 \equiv E_d$ modulo both prime powers, E_d is idempotent modulo 10^d . The residues are compatible as d grows.

Theorem 8.1 (All initial-length tails and their signs). *For $n \geq 3$ and $1 \leq d \leq n$,*

$$\rho(n) \equiv \delta_n E_d \pmod{10^d}, \quad \delta_n = \begin{cases} -1, & n \equiv 1 \pmod{4}, \\ +1, & n \not\equiv 1 \pmod{4}. \end{cases} \quad (30)$$

In particular, for every $n \geq 10$ the last ten digits of $\rho(n)$ are

$$\begin{cases} 1787109375, & n \equiv 1 \pmod{4}, \\ 8212890625, & n \not\equiv 1 \pmod{4}. \end{cases} \quad (31)$$

Proof. Because $5 \mid r(n)$, $5^n \mid \rho(n)$. If n is even, apply (8) to $r(n)^n$:

$$\nu_2(\rho(n) - 1) = c_2(r(n)) + \nu_2(n) = s + \nu_2(n) = n.$$

Hence $\rho(n) \equiv 1 \pmod{2^n}$. If n is odd, $s = n$ and (5) becomes

$$r(n) = \begin{cases} 3 \cdot 2^n - 1, & n \equiv 1 \pmod{4}, \\ 3 \cdot 2^n + 1, & n \equiv 3 \pmod{4}. \end{cases}$$

Raising to the odd exponent n preserves the indicated sign modulo 2^n . Thus in all cases $\rho(n) \equiv \delta_n \pmod{2^n}$. Apply the Chinese remainder theorem and reduce to any $d \leq n$. Finally, direct calculation from (29) gives $E_{10} = 8212890625$ and $10^{10} - E_{10} = 1787109375$. $\square$

The ten-digit assertion is stronger than simply choosing one of two possible suffixes: it determines which suffix occurs from $n \bmod 4$. It also starts at $n = 10$, one index earlier than the range $n > 10$ specified in the question [4].

There are two distinct limiting notions in this discussion. The compatible residues E_d describe a 10-adic integer with components $(1, 0)$ in $\mathbb{Z}_2 \times \mathbb{Z}_5$. They are not a claim that the ordinary real sequence $\rho(n)$ converges. They describe the minimizing *bases*, not the eventual tail of a tower formed from one fixed base.

9 A two-parameter extremal theorem

The proof is not confined to the diagonal where degree and speed coincide. For positive integers k, m , put

$$R_k(m) = \min\{x > 1 : 10 \nmid x, V(x^k) = m\}$$

whenever this set is nonempty. Here k is the power degree and m is the requested speed.

Theorem 9.1 (An explicit region of global minimality). *Let*

$$s = m - \nu_2(k) \geq 2, \quad t = m - \nu_5(k) \geq 1.$$

If

$$5^t > B_s^2 + 1, \tag{32}$$

then $R_k(m) = B_s$, and the least k th perfect power of speed m is B_s^k .

Proof. By (16), B_s has exactly the requested speed after raising to the k th power. It is least among roots divisible by 5. Any competing root not divisible by 5 has $c_5(x) \geq t$, so Lemma 5.1 and (32) rule it out. $\square$

Corollary 9.2 (Fixed-degree eventual formula). *For every fixed k , the equality*

$$R_k(m) = B_{m-\nu_2(k)}$$

holds for all sufficiently large m . A directly checkable sufficient condition, in addition to $s \geq 2, t \geq 1$, is

$$\left(\frac{5}{4}\right)^m \geq 16 \frac{5^{\nu_5(k)}}{4^{\nu_2(k)}}. \tag{33}$$

Proof. For $s \geq 2$,

$$B_s^2 + 1 \leq (3 \cdot 2^s + 1)^2 + 1 < 16 \cdot 4^s.$$

Condition (33) gives $5^{m-\nu_5(k)} \geq 16 \cdot 4^{m-\nu_2(k)} > B_s^2 + 1$. For fixed k , the left side of (33) tends to infinity, while the right side is constant. $\square$

This result gives an effective answer in an infinite region of the (k, m) plane. It does not assert that the displayed inequality is necessary, nor does it claim a simple formula for every small pair outside that region.

10 Exact densities and what the heuristic was seeing

A density argument is not a minimality proof: a sparse set can still contain the smallest element. Nevertheless, the local formulas make precise the statistical preference for roots divisible by 5. For a subset $A \subseteq \mathbb{N}$, its natural density, when it exists, is

$$\lim_{X \rightarrow \infty} \frac{\#(A \cap [1, X])}{X}.$$

All densities below exist because the conditions, apart from the single excluded integer 1, are finite unions of residue classes.

Theorem 10.1 (Tail and exact-speed densities). *Fix $k, m \geq 1$ with $s = m - \nu_2(k) \geq 2$ and $t = m - \nu_5(k) \geq 1$. The density, among all positive integers x , of*

$$\{x > 1 : 10 \nmid x, V(x^k) \geq m\}$$

is

$$\mathcal{D}_k(m) = \frac{2^{1-s}}{5} + \frac{2}{5^t} + \frac{2^{3-s}}{5^t}. \quad (34)$$

The density of the corresponding set with $V(x^k) = m$ is

$$\mathcal{E}_k(m) = \frac{2^{-s}}{5} + \frac{8}{5^{t+1}} + \frac{9 \cdot 2^{2-s}}{5^{t+1}}. \quad (35)$$

Proof. For odd x , $c_2(x) \geq s$ is equivalent to $x \equiv \pm 1 \pmod{2^s}$, which occupies $2/2^s$ of all integers. For $5 \nmid x$, $c_5(x) \geq t$ is equivalent to $x^4 \equiv 1 \pmod{5^t}$. There are exactly four roots: 1, -1 , and the two roots of -1 from Lemma 6.1.

Partition the permitted integers into three disjoint types. Odd multiples of 5 contribute $(2/2^s)(1/5) = 2^{1-s}/5$. Even integers not divisible by 5 contribute $(1/2)(4/5^t) = 2/5^t$. Integers coprime to 10 contribute $(2/2^s)(4/5^t) = 2^{3-s}/5^t$. The products follow from the Chinese remainder theorem. Their sum proves (34). For exact speed, subtract the density at threshold $m+1$, which replaces s, t by $s+1, t+1$. This gives (35). $\square$

These are densities among *roots* x , not among the much sparser set of k th powers as numbers on the ordinary number line. Also the denominator includes all positive integers; conditioning on $10 \nmid x$ would divide the displayed densities by $9/10$.

For the diagonal $k = m = n$, the contribution $2^{-s}/5$ to exact-speed density comes from odd multiples of 5. The ratio of the other two contributions to this one is exactly

$$\frac{8 \cdot 2^s + 36}{5^t}.$$

Since $2^s \leq 2^n$ and $5^t \geq 5^n/n$, this ratio tends to zero. Thus almost all eligible roots, in the density sense as the parameter n grows, belong to the final-digit-5 class. This explains the heuristic but is logically separate from the proof that the *least* root always belongs to that class for $n \geq 3$.

11 Asymptotics and oscillations of the minimum

Write

$$c_s = \begin{cases} 1, & s \text{ even}, \\ 3, & s \text{ odd}, \end{cases} \quad \varepsilon_s = \begin{cases} -1, & s \equiv 0, 1 \pmod{4}, \\ +1, & s \equiv 2, 3 \pmod{4}. \end{cases}$$

Then $r(n) = c_s 2^s + \varepsilon_s$ for $n \geq 3$.

Proposition 11.1 (Logarithmic asymptotics). *As $n \rightarrow \infty$,*

$$\log \rho(n) = n(n - \nu_2(n)) \log 2 + n \log c_s + O(n 2^{-s}). \quad (36)$$

Consequently

$$\lim_{n \rightarrow \infty} \rho(n)^{1/n^2} = 2, \quad \log_2 \rho(n) = n^2 + O(n \log n). \quad (37)$$

Proof. Factor $r(n) = c_s 2^s (1 + \varepsilon_s / (c_s 2^s))$ and use $\log(1+u) = O(u)$ uniformly for $|u| \leq 1/4$. Multiply by n . Since $\nu_2(n) \leq \log_2 n$, we have $s \geq n - \log_2 n \rightarrow \infty$. Dividing (36) by n^2 then proves both conclusions. $\square$

Despite the clean quadratic logarithmic growth, the root sequence is not monotone. Some adjacent values coincide; others decrease sharply. For example, if $n = 4j + 1 \geq 5$, then

$$n - \nu_2(n) = n = (n + 1) - \nu_2(n + 1),$$

so $r(4j + 1) = r(4j + 2)$.

Proposition 11.2 (Complete accumulation set of normalized roots). *The set of subsequential limits of $r(n)/2^n$ is*

$$\{0, 3\} \cup \{3/2^j : j \geq 1 \text{ odd}\} \cup \{1/2^j : j \geq 2 \text{ even}\}. \quad (38)$$

In particular, $\liminf r(n)/2^n = 0$ and $\limsup r(n)/2^n = 3$.

Proof. Along odd n , $s = n$ is odd and the normalized root tends to 3. Along integers with fixed $j = \nu_2(n) \geq 1$, the integer n is even, so $s = n - j$ is odd precisely when j is odd. Hence

$$\frac{r(n)}{2^n} = \frac{c_s}{2^j} + O(2^{-n})$$

has the indicated limit. Each fixed valuation class contains infinitely many integers. Along $n = 2^j$ with $j \rightarrow \infty$, the ratio tends to zero. Conversely, along a subsequence of indices tending to infinity, either the 2-adic valuations have an unbounded subsequence, forcing a further limit of zero, or some fixed valuation occurs infinitely often. If the original subsequence converges, its limit must be one of these values. $\square$

The limit $\rho(n)^{1/n^2} \rightarrow 2$ concerns a sequence of specially chosen *bases*. It should not be confused with a bound on the growth of $t \mapsto a \uparrow\uparrow t$, which is a different variable and a very different sequence.

12 Algorithms, validation, and reproducibility

12.1 The constant-size formula is not a tower computation

Once n is given, the formula requires the valuation $\nu_2(n)$, a residue modulo 4, a binary shift, and one small multiplication and addition. The full root $r(n)$ has $\Theta(n)$ bits, while $\rho(n)$ has $\Theta(n^2)$ bits. Thus writing the root or the power has an unavoidable output cost. Describing the formula as “constant time” would be misleading when the input n is encoded in binary.

A direct implementation of the decisive part is:

```
def minimum_root(n: int) -> int:
    if n < 1:
        raise ValueError("n must be positive")
    if n == 1:
        return 2
    if n == 2:
        return 7
    v2 = (n & -n).bit_length() - 1
    s = n - v2
    coefficient = 3 if s % 2 else 1
    sign = -1 if s % 4 in (0, 1) else 1
    return coefficient * (1 << s) + sign
```

The accompanying module includes argument validation, speed evaluation, modular tails, the exact-degree variant, and command-line support. It uses Python's standard library only. To compute the full answer, one evaluates `pow(minimum_root(n), n)`. Repeated squaring uses $O(\log n)$ integer multiplications; their sizes must be included in a bit-complexity estimate.

For only the last $d \leq n$ decimal digits with $n \geq 3$, use Theorem 8.1; it does not require materializing a root of $\Theta(n)$ bits. Formula (29) involves integers with $O(d)$ bits and one modular inverse, followed by the sign test on $n \bmod 4$.

12.2 What was actually verified

The included run was executed with Python 3.13.5. All tests passed. The detailed machine-readable record is `data/verification.json`. The checks deliberately use more than one computational route.

First, the compact speed formula was compared with a separate transcription of the published residue-class formula for every $a = 2, \dots, 20000$ not divisible by 10: 17,999 comparisons. Second, the power-transfer formula was checked against actual integer powers and that separate formula for $2 \leq x \leq 500$, $10 \nmid x$, and $1 \leq k \leq 30$: 13,470 comparisons.

Third, unrestricted minima were exhaustively searched for $1 \leq n \leq 20$, using actual values x^n rather than merely checking the proposed root. Exact-degree minima were exhaustively searched for $1 \leq n \leq 15$. Every proposed root was independently checked to have speed n through $n = 10000$. The first 50 roots agree with the values reported by Alekseyev in the source discussion.

Fourth, the integer comparisons in Table 1 and the Hensel certificates in Table 2 were regenerated exactly. Modular tail identities were checked for $3 \leq n \leq 2000$ at lengths $1, \min(n, 10), \min(n, 30)$. A separate perfect-power test checked the assertion about B_s for $2 \leq s \leq 200$.

Finally, tower residues were computed by an independent modular exponentiation routine for 17 representative bases, through tower height 14, with 160 decimal digits of modulus. The resulting $D_0, \dots, D_{13}$ agree with the predicted eventual speeds. The tower routine itself was first checked in 88 comparisons with literal towers small enough to construct. Several complete residue periods were also enumerated to validate the density formula.

The role of these computations is verification and error detection. The infinite quantifier in Theorem 1.2 is discharged by Lemma 5.3, not by checking roots through $n = 10000$.

12.3 Reproduction commands

From the archive's top-level directory, run

```
python code/verify.py
python code/tetration_minima.py 15 --root-only
python code/tetration_minima.py 3 --strict
python code/tetration_minima.py 100 --digits 10
python code/tetration_minima.py --table roots.csv --count 1000
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The second command prints 98305; the third prints 166375. The fourth prints 8212890625. Running the verification script regenerates the CSV files and verification record, including the finite certificates. No network access, symbolic algebra system, or floating-point computation

is required by the mathematical checks. Elapsed time in the log is descriptive of that particular run, not a portable performance guarantee.

13 What has been settled and what has not

For the definitions in Section 1, the global minimum is now given by (5) for every positive n . The proof also determines the maximal-degree variant and the precise ten-digit-tail choice. The unrestricted minimum and its exact-degree strengthening are not conditional on a conjecture or an unverified random-digit assumption.

The historical claim is narrower: the specific conjectural pattern in the consulted MathOverflow question and answer is proved here. The underlying speed formula, the method of using valuations, the observed initial values, and the idea that the final-digit-5 branch should win all have prior provenance. This report's contribution is the explicit global argument based on the quadratic barrier, together with the closed formula and the derived refinements. The sources checked are not an exhaustive review of all unpublished notes or discussions. Independent review remains appropriate before making a formal priority claim or submitting the result for publication.

The two-parameter theorem leaves small pairs outside its sufficient region untreated, and no arbitrary-radix analogue is proved. A natural next problem is to determine which radices admit a similarly simple extremal branch. The crucial feature here is that a 5-adic fourth-root condition factors into pieces of degree at most two, making a $5^{m/2}$ size barrier available. In another radix, the relevant cyclotomic factors and competing prime-adic slopes can change the comparison. Such an extension would require its own proof rather than a formal replacement of the number 10.

A A short proof suitable for checking independently

For a reader willing to use the known congruence-speed formula, the entire main argument compresses to the following. For odd $x > 1$, put $c_2(x) = \nu_2(x^2 - 1) - 1$; for $5 \nmid x > 1$, put $c_5(x) = \nu_5(x^4 - 1)$. Lifting gives $c_2(x^n) = c_2(x) + \nu_2(n)$ and $c_5(x^n) = c_5(x) + \nu_5(n)$.

Set $s = n - \nu_2(n)$ and $t = n - \nu_5(n)$. The least odd multiple of 5 with $c_2(x) = s \geq 2$ is B_s , by solving $x = j2^s \pm 1$ with j odd and then reducing modulo 5. It yields $V(B_s^n) = n$.

Any other competing root must satisfy $c_5(x) \geq t$. Since exactly one of $x - 1, x + 1, x^2 + 1$ is divisible by 5, it follows that $x^2 + 1 \geq 5^t$. Hence $5^t > B_s^2 + 1$ proves global minimality. For $n \geq 25$, use

$$5^t \geq 5^n/n > (3 \cdot 2^n + 1)^2 + 1 \geq B_s^2 + 1;$$

the middle inequality follows from its checked value at 25 and the ratio argument in Lemma 5.3.

For $3 \leq n \leq 24$, the same inequality fails only at 3, 5, 7, 9, 10, 15. At those indices, use respectively the smaller roots 57, 182, 1068, 32318, 32318, 110443 of -1 modulo $5^3, 5^4, 5^5, 5^7, 5^7, 5^8$. They and the least nontrivial ± 1 residues exceed B_s . This excludes every remaining competitor. Checking $n = 1, 2$ gives 2 and 7.

This summary uses only finitely many small, explicit arithmetic certificates. The longer article proves the speed formula and the lifting steps as well.

B Tables and archive inventory

B.1 First twenty-five minimizing roots

n	$r(n)$	n	$r(n)$	n	$r(n)$
1	2	10	1535	19	1572865
2	7	11	6145	20	262145
3	25	12	1025	21	6291455
4	5	13	24575	22	6291455
5	95	14	24575	23	25165825
6	95	15	98305	24	6291455
7	385	16	4095	25	100663295
8	95	17	393215		
9	1535	18	393215		

For the exact-degree sequence, replace only the root at $n = 3$ by 55. The desired base is the n th power of the root in this table.

B.2 A lifting chain for square roots of -1

The following chain starts at 2 (mod 5) and follows its unique lift. The other root is the negative modulo 5^k .

k	5^k	lift of 2 (mod 5)	smaller root d_k
1	5	2	2
2	25	7	7
3	125	57	57
4	625	182	182
5	3125	2057	1068
6	15625	14557	1068
7	78125	45807	32318
8	390625	280182	110443

B.3 Included files

article.tex, article.pdf

The complete source and rendered article.

code/tetration_minima.py

The exact formulas, speed functions, Hensel lifts, tails, strict variant, and command-line interface.

code/verify.py

Independent exhaustive checks and generation of all finite certificates.

data/minimum_roots_1_1000.csv

The first 1000 roots, their valuation parameters, and checked speeds.

data/rho_1_50.txt

The first 50 complete integers $\rho(n)$ in index-value format.

data/finite_certificates.csv, data/small_range.csv

Machine-readable versions of the finite proof calculations.

data/verification.json, data/verification.log

The validation results, including independent tower checks.

README.md, **sources.md**, **build.sh**

Usage notes, source provenance and status, and a simple build command.

References

- [1] M. Ripà, *On the constant congruence speed of tetration*, Notes on Number Theory and Discrete Mathematics **26** (2020), no. 3, 245–260. [doi:10.7546/nntdm.2020.26.3.245-260](https://doi.org/10.7546/nntdm.2020.26.3.245-260).
- [2] M. Ripà, *The congruence speed formula*, Notes on Number Theory and Discrete Mathematics **27** (2021), no. 4, 43–61. [doi:10.7546/nntdm.2021.27.4.43-61](https://doi.org/10.7546/nntdm.2021.27.4.43-61). Also [arXiv:2208.02622](https://arxiv.org/abs/2208.02622) (2022).
- [3] M. Ripà and L. Onnis, *Number of stable digits of any integer tetration*, Notes on Number Theory and Discrete Mathematics **28** (2022), no. 3, 441–457. [doi:10.7546/nntdm.2022.28.3.441-457](https://doi.org/10.7546/nntdm.2022.28.3.441-457).
- [4] M. Ripà (question) and M. Alekseyev (answer), *Closed form for the general term of 2, 49, 15625, 625, ...*, MathOverflow, question 487698, 12–14 February 2025. <https://mathoverflow.net/questions/487698/>. Question, answer, and associated conjectural statements checked 19 September 2026.
- [5] M. Ripà, *On the relation between perfect powers and tetration frozen digits*, Journal of AppliedMath **2** (2024), no. 5, Article 1771. [doi:10.59400/jam1771](https://doi.org/10.59400/jam1771). Also [arXiv:2602.00252v1](https://arxiv.org/abs/2602.00252v1), 30 January 2026. Cited for the existence program and context, not as a proof of the extremal theorem established here.